

Kun Pang (Hendrix)

Incoming Ph.D. Student · ECE · University of Virginia (Fall 2026) | Advised by Prof. Steven Bowers

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Profile

Strong foundation in RF/electromagnetics and communication systems, transitioning toward **RFIC design and integrated photonics**. Hands-on experience in mm-wave simulation, antenna/metasurface design, and SDR implementation; currently building circuit- and photonic-level foundations to bridge system/EM intuition with IC and photonic design.

Education

University of Virginia, Charlottesville, VA, USA Fall 2026 (Incoming)

Ph.D. in Electrical & Computer Engineering — Advised by Prof. Steven Bowers

Sun Yat-sen University, Guangzhou, China 2022 – 2026 (Expected)

B.Eng. in Information Engineering (Elite Class) GPA: 3.7/4.0

Core RF/Comm courses (DSP, Communication Theory, Radar, Automatic Control): 4.0/4.0 | TOEFL: 104

Research Experience

Reconfigurable Intelligent Surfaces (RIS) for mm-Wave Systems Dec 2024 – Dec 2025

Prof. Shaoqiu Xiao, Sun Yat-sen University

- Designed K-band (20–26 GHz) PIN-diode RIS unit cells: $\sim 360^\circ$ phase tuning, < 1.6 dB loss, near-ideal linearity ($R^2 = 0.9991$) in HFSS. **First-author paper at EuCAP 2026** (Dublin).

Microstrip Antenna Array Design Mar 2024 – Dec 2024

Prof. Shaoqiu Xiao, Sun Yat-sen University

- 4-element patch array at 5.8 GHz (< -20 dB isolation) for MIMO; validated via anechoic-chamber & S-parameter measurements.

Selected Projects

Dual Circularly-Polarized Satellite Antenna (13–14.5 GHz) Oct 2025 – Present

Prof. Shaoqiu Xiao, Sun Yat-sen University

- Multilayer dual-CP antenna with isolation-enhanced structure; optimization, simulation, and testing.

BPSK Digital Voice Transmission System (SDR) May 2025

Lingte Cup National RF Competition — First Prize

- End-to-end voice link on PlutoSDR + GNU Radio: channel coding, BPSK modulation, equalization ($\text{BER} < 10^{-6}$).

Publications

- K. Pang**, Y. Chen, H. Lu, B. Lei, S. Xiao. “Enhanced Linear Phase Control of a K-Band RIS via Coupled Hollow I-Shaped Resonators.” *EuCAP 2026*, Dublin.
- B. Lei, ..., **K. Pang**, et al. “Radiation performance analysis of asymmetric-structured self-biased magnetoelectric antennas via multi-field coupling modeling.” *Smart Materials and Structures*, 2025. DOI:10.1088/1361-665X/ae1beb
- B. Lei, ..., **K. Pang**, et al. “Theoretical analysis on radiation performance of magnetoelectric antennas considering eddy-current loss and quasistatic hysteresis effect.” *J. Appl. Phys.*, 138(4), 044101, 2025. DOI:10.1063/5.0282234

Currently Building (RFIC / Photonics Transition)

- RFIC**: self-studying Razavi *RF Microelectronics* (nonlinearity, noise, phase noise); Cadence Virtuoso/Spectre (in progress).
- Foundations**: control theory, stochastic processes, probability for noise analysis; integrated-photonics device concepts.

Technical Skills

- RF/EM & Circuit**: HFSS, ADS, VNA, anechoic-chamber measurement; Cadence Virtuoso/Spectre (in progress)
- SDR & Comm**: GNU Radio, PlutoSDR; BPSK/QPSK/QAM, OFDM, MIMO, channel coding | **Programming**: MATLAB/Simulink, Python, C++, Qt5, Git

Research Interests

RFIC Design · Integrated Photonics · Millimeter-wave Systems · RF/Analog Circuits · Wireless Communications